

Modeling Legal Contracts

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Abstract

Recall that a legal contract has both a “logical component” (obligations, permissions and prohibitions) and a payoff (“performance”) structure. The purpose of this document is to give a precise, mathematical definition of the logical component for a certain class of legal contracts, using tools from mathematical logic and philosophy. We begin with a small language (“classical logic”) and we extend it to a language capable of defining a meaningful class of legal contract.

1 Propositional Logic

Let Σ be a countably infinite set of symbols. Consider the connectives

$$\wedge, \vee, \neg, \rightarrow$$

each of which can be defined through an obvious truth table. We define the language L by inductively building sentences with the connectives. This is classical propositional logic.

We also recall that we have a “turnstile” symbol $\vdash$, which is not part of propositional logic. Instead, it belongs to a meta-language. It works as follows. If P is a set of premises, and one can derive Q from P (using logic and direct manipulation), then we say

$$P \vdash Q.$$

If we have no premises, then

$$\vdash Q$$

means that Q follows from the axioms of the system with **no assumptions**.

2 Deontic Logic

We now give an informal introduction to deontic logic.

For a proposition p , we have the following statements:

1. $\mathcal{O}p$: it is **obligatory** that p
2. $\mathcal{P}p$: it is **permitted** that p
3. $\mathcal{F}p$: it is **forbidden** that p

Now assume that the operator $\mathcal{O}$ is already defined. We define

$$\mathcal{P}p := \neg\mathcal{O}\neg p$$

and

$$\mathcal{F}p := \mathcal{O}\neg p.$$

We have the following axioms

1. **K** (distributive) :

$$\mathcal{O}(p \rightarrow q) \rightarrow (\mathcal{O}(p) \rightarrow \mathcal{O}(q))$$

2. **D** (serial) :

$$\mathcal{O}p \rightarrow \neg\mathcal{O}\neg p$$

3. **N** (necessity) :

$$\vdash p \rightarrow \vdash \mathcal{O}p$$

This completes the definition of SDL as a formal language.

3 Kripke Models

Recall that for a set X , 2^X denotes its power set, i.e. the set of subsets of X . A Kripke model $\mathcal{M}$ consists of the following three pieces of data

- A set of worlds W
- A relation $R \subset W \times W$
- A function (called a **valuation**)

$$V : \Sigma \rightarrow 2^W \quad V(p) = \{w \in W : p \text{ is true in } w\}.$$

For a world w and a proposition p , we say w **models** p , denoted by $w \models p$ if $w \in V(p)$. We extend the definition of $\models$ on atomic propositions as follows:

$$\begin{aligned} w \models \neg p &\iff \neg(w \models p) \\ w \models (p \wedge q) &\iff (w \models p) \wedge (w \models q) \\ w \models (p \vee q) &\iff (w \models p) \vee (w \models q) \\ w \models (p \rightarrow q) &\iff (w \models p) \rightarrow (w \models q) \end{aligned}$$

We now define it on SDL sentences. We say

$$w \models \mathcal{O}p \iff \forall w \in W : wRw \rightarrow w \models p$$

and

$$w \models \mathcal{P}p \iff \exists v \in W : wRv \wedge v \models p.$$

4 Dyadics

Recall the notion of **order** in mathematics, and let us consider orders on W . We say a relation $\preceq \subset W \times W$ is a **partial order** if it is both reflexive and transitive. We say $\preceq$ is a **total order** if it is total, i.e. for any $w_1, w_2 \in W$, $w_1 \preceq w_2$ or $w_2 \preceq w_1$, and antisymmetric, i.e. if $w_1 \preceq w_2$ and $w_2 \preceq w_1$ then $w_1 = w_2$.

This distinction is important. Let $\preceq$ be a partial order on W . Then if $U \subset W$, then we say $x \in U$ is **minimal** if $x \preceq u$ for all $u \in U$. If the order is total, then this element is unique. If it is not total, then it is possibly non-unique. We denote the set of minima by

$$\min_{\preceq} U.$$

Now assume that for each world $w \in W$ we have a partial order $\preceq_w \subset W \times W$ relative to w . Then we consider

$$[p] = \{w \in W : w \models p\}$$

and we define

$$w \models \mathcal{O}(p|q) \iff \min_{\preceq_w} [q] \subset [p].$$

We say $(M, W, \preceq, V)$ is a **dyadic deontic model**.

5 Simple Contracts

We are now ready to define a simple class of legal contract. Let $(M, W, \preceq, V)$ be a dyadic deontic model. We define a **contract** as

$$\mathcal{C} = \{\mathcal{O}(p_1|q_1), \dots, \mathcal{O}(p_n|q_n)\}$$

and say $\mathcal{C}$ is **satisfied** with respect to $w \in W$ if $M, w \models \phi$ for all $\phi \in \mathcal{C}$. We denote this by

$$M, w \models \mathcal{C}.$$